

Student Name: <u>Jacky Shen</u>	Final Examination Part I
Date: Monday May-04, 2026	Time: 01:00pm-03:00pm

(19) + (50) = 69

Question #01 (5pts): Select the correct answer for the Gibbs free energy G^{ID} of an ideal liquid mixture of two components with molar composition $\{x_1, x_2\}$.

☐	$G^{ID} = x_1 \bar{G}_1 + x_2 \bar{G}_2$
☐	$G^{ID} = x_1 \bar{G}_1^{ID} + x_2 \bar{G}_2^{ID}$
☒	$G^{ID} = x_1 \bar{G}_1^{ID} + x_2 \bar{G}_2^{ID} + RT(x_1 \ln x_1 + x_2 \ln x_2)$

Question #02 (5pts): Select the correct answer for the definition of the K -values of an ideal gas mixture in equilibrium with a non-ideal liquid mixture.

☒	$K_i = \frac{\hat{\phi}_i^V}{\hat{\phi}_i^L}$
☐	$K_i = \frac{P_i^{\text{sat}}}{P}$
☐	$K_i = \hat{\phi}_i^L$

Question #03 (5pts): Select the correct answer for the equation of dew point temperature of a non-ideal vapor mixture with molar composition $\{y_1, y_2, \dots, y_N\}$. Assume ideal-liquid phase mixture

☒	$\sum_{i=1}^N y_i \hat{\phi}_i^V = 1$
☐	$\sum_{i=1}^N y_i \hat{\phi}_i^L = 1$
☐	$\sum_{i=1}^N y_i \frac{\hat{\phi}_i^V}{\hat{\phi}_i^L} = 1$
☐	$\sum_{i=1}^N y_i \frac{\phi_i^L}{\hat{\phi}_i^V} = 1$

what is the pressure at the beginning and the end of the vaporization process. Saturated vapor pressures of benzene and toluene are 1008 mmHg and 404 mmHg respectively at 90 °C

Pressure at the beginning (mmHg)	404
Pressure at the end (mmHg)	1008

2

Question #05 (5pts): The following Equation of State $\frac{PV}{RT} = 1 + \frac{b}{V}$, where b is a constant number, describes the properties of a pure fluid. Derive a formula for the Residual Entropy as a function of (Z, B) where $B = \frac{bP}{RT}$

$\frac{S^R}{R} =$	0
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Question #06 (5pts): The Equilibrium constant of a chemical reaction is independent of the pressure of the reaction mixture.

☒	True
☐	False

5

Question #07 (5pts): An ideal gas expands **isenthalpically** from a pressure $P_1 = 10 \text{ bar}$ to a pressure $P_2 = 1 \text{ bar}$. What is the temperature change of the gas during this process (select from the list below)

☒	$\Delta T = 0$
☐	$\Delta T < 0$
☐	$\Delta T > 0$

5

Question #08 (5pts): Consider the total differential $d(nG) = nVdP - nSdT + \sum_i \bar{G}_i dn_i$. Which formulas from the ones shown below are correct. You must select all that apply for full credit

☐	$\left(\frac{\partial S}{\partial P} \right)_{T,n} = - \left(\frac{\partial V}{\partial T} \right)_{P,n}$
☒	$\left(\frac{\partial \bar{G}_i}{\partial P} \right)_{T,n_j, j \neq i} = \bar{V}_i$
☒	$\left(\frac{\partial \bar{G}_i}{\partial T} \right)_{P,n_j, j \neq i} = -\bar{S}_i$

2

Question #09 (5pts): The fugacity coefficient of a component inside an ideal-liquid mixture is equal to one.

☐	True
☒	False

5

Question #10 (5pts): A single VLE experimental data point is available at the bubble point temperature $T_{BUBBLE} = 35^\circ\text{C}$ under pressure $P = 76.719\text{ kPa}$ with composition $[x_1, y_1] = [0.1665, 0.0832]$ for a binary mixture. The saturated vapor pressures of the two components are $P_1^{\text{sat}} = 49.624\text{ kPa}$ and $P_2^{\text{sat}} = 85.265\text{ kPa}$ at the bubble point temperature. What is the value of the coefficient A in the Margules 2-suffix equation for the Excess Gibbs Free Energy. Assume ideal gas mixture for the vapor phase.

Value of A

x_1	x_2	y_1	y_2	K_1	K_2
0.1665	0.8335	0.0832	0.9168	$\frac{49.624}{76.716}$	$\frac{85.265}{76.716}$
				0.647	1.111

$K = \frac{P_i^{\text{sat}}}{P}$

Bubble point, X

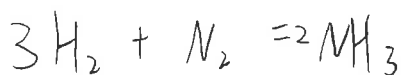
RR equation will be depending on the K values and an α of 1 for liquid phase

Student Name: <u>Jacky Shen</u>	Final Examination Part II
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Problem #01 (15 pts): A gas mixture containing 70 mol% Hydrogen, 20 mol% Nitrogen and 10 mol% inert gas Argon is passed over a catalyst for production of Ammonia. If the pressure of the reaction mixture is 50 atm and the final temperature 400 °C, calculate the percentage of Hydrogen converted to Ammonia and the mole fraction of Ammonia in the reaction mixture. Assume the gas mixture follows the Peng-Robinson Equation of State with Binary Interaction Parameters all set equal to zero.

H ₂ Conversion (%)	25.26%
NH ₃ mole fraction	0.1337

15



400 °C

$$K_{eq} \text{ Gibbs} = 0.0002$$

$$x_{\text{H}_2} = \frac{3\varepsilon}{0.7} = 0.2526 = \boxed{25.26\%}$$

$$y_{\text{NH}_3} = \frac{0.117878}{0.882122} = \boxed{0.1337}$$

$$\varepsilon = 0.0589391$$

$$n_{\text{H}_2} = 0.7 - 3\varepsilon = 0.523183$$

$$n_{\text{N}_2} = 0.141061$$

$$n_{\text{Ar}} = 0.1$$

$$n_{\text{NH}_3} = 0.117878$$

$$n_T = 0.882122$$

Problem #02 (15 pts): The following Equation of State has been proposed for the calculation of the thermal properties of Nitrogen:

$$P + \frac{a}{TV^2} = \frac{RT}{V-b} \quad \left| \quad a = \lambda_1 \frac{R^2 T_c^3}{P_c} \quad \right| \quad b = \lambda_2 \frac{RT_c}{P_c}$$

Calculate the values of the numerical constants $\{\lambda_1, \lambda_2\}$ and the value of the critical compressibility factor Z_c . Derive an expression for the calculation of the residual enthalpy and calculate its value at $T=50^\circ\text{C}$ and pressure $P=650\text{ atm}$. **Hint:** Use an initial guess of $Z^{(0)} = 2$ for the calculation of the residual enthalpy

$$P + \frac{a}{TV^2} = \frac{RT}{V-b} \quad \left. \frac{\partial P}{\partial V} \right|_T = 0$$

$$P = \frac{RT}{V-b} - \frac{a}{TV^2} \quad \left. \frac{\partial^2 P}{\partial V^2} \right|_T = 0$$

$$\frac{\partial P}{\partial V} = -\frac{RT}{(V-b)^2} + \frac{2a}{TV^3} = 0 \quad \text{eq 1}$$

$$\frac{\partial^2 P}{\partial V^2} = \frac{2RT}{(V-b)^3} - \frac{6a}{TV^4} = 0 \quad \text{eq 2}$$

$$\text{eq 2 / eq 1} = 3b = V_c$$

$$P_c = \frac{RT_c}{3b-b} - \frac{a}{T_c(3b)^2}$$

$$P_c = \frac{RT_c}{2b} - \frac{a}{9T_c b^2}$$

$$a = \frac{27}{8} RT_c^2 b$$

$$P_c = \frac{RT_c}{8b}$$

$$b = \frac{RT_c}{8P_c} = \lambda_2 \frac{RT_c}{P_c}$$

$$\lambda_2 = \frac{1}{8} = 0.125$$

$$Z_c = \frac{P_c V_c}{RT_c}$$

$$V_c = 3b = 3 \left(\frac{RT_c}{8P_c} \right)$$

$$Z_c = \frac{P_c}{RT_c} \left(\frac{3RT_c}{8P_c} \right)$$

$$Z_c = \frac{3}{8} = 0.375$$

$$H^R = RT(Z-1) + \int_{\infty}^V \left[T \left(\frac{\partial P}{\partial T} \right)_V - P \right] dV$$

$$T \left(\frac{\partial P}{\partial T} \right)_V - P = \frac{2a}{TV^2}$$

$$H^R = RT(Z-1) + \int_{\infty}^V \frac{2a}{TV^2} dV = RT(Z-1) - \frac{2a}{RT^2 V}$$

$$H^R = 7.493 \frac{\text{Latm}}{\text{mol}} = 759.232 \frac{\text{J}}{\text{mol}}$$

$$\frac{H^R}{RT} = (Z-1) - \frac{2a}{RT^2 V}$$

$$a = \frac{27}{8} RT_c^2 \left(\frac{RT_c}{8P_c} \right) = \frac{27}{64} \frac{R^2 T_c^3}{P_c} = \lambda_1 \frac{R^2 T_c^3}{P_c}$$

$$\lambda_1 = \frac{27}{64} = 0.421875$$

Using $T_c = -146.95^\circ\text{C}$

$$P_c = 33.5\text{ atm}$$

$$T = 50^\circ\text{C}$$

$$P = 650\text{ atm}$$

λ_1	0.421875
λ_2	0.125
Z_c	0.375
$\frac{H^R}{RT}$	$(Z-1) - \frac{2a}{RT^2 V}$

15

Problem #03 (15 pts): The mixture of Methanol (Component #1), Acetone (Component #2) and Cyclohexane (Component #3) at $T = 101^\circ\text{C}$ forms an azeotrope $\{y_1 = x_1, y_2 = x_2 \text{ and } y_3 = x_3\}$. Determine the pressure and composition that is predicted for the azeotrope between methanol, acetone and cyclohexane by using the NRTL method with binary interaction parameters, $\{A_{ij}, A_{ji}, B_{ij}, B_{ji}, C\}$ shown in the table and assuming that the vapor phase follows the **Redlich-Kwong** Equation of State with all BIPs set equal to zero. **Poynting Factor=1**

BIPs		CH ₄ O C ₃ H ₆ O	CH ₄ O C ₆ H ₁₂	C ₃ H ₆ O C ₆ H ₁₂
K_{Aij}		0.0000	0.0000	0.0000
K_{Bij}				
K_{Cij}				
A_{ij}		0.0000	1.3869	-0.1061
A_{ji}		0.0000	-4.6753	-3.5142
B_{ij}		114.1347	224.7601	359.6793
B_{ji}		101.8859	2277.7942	1485.3369
C		0.3000	0.4300	0.4700

Pressure (bar)	4.9480
Methanol mole fraction	0.6034
Acetone mole fraction	0.1115
Cyclohexane mole fraction	0.2851

15

$$p = \frac{\sum_i x_i \phi_i^{\text{sat}} P_i^{\text{sat}}}{\phi_i}$$

Problem #04 (15 pts): A liquid hydrocarbon mixture of Methane (Comp #1), Ethane (Comp #2) and Propane (Comp #3) at temperature $T_1 = -100\text{ }^{\circ}\text{C}$ and pressure $P_1 = 64\text{ bar}$ with molar composition $z = [0.80, 0.08, 0.12]$ expands **adiabatically** down to a pressure of $P_2 = 10\text{ bar}$ in a liquid turbine. If the temperature at the outlet of the liquid turbine is $T_2 = -118.5\text{ }^{\circ}\text{C}$ calculate the power generated in kW. The flow rate of the hydrocarbon mixture is 10,000 kmol/hr. Also report the calculated vapor fraction at the turbine outlet and the Lost Work in kW. Assume that both phases of the hydrocarbon mixture follow the Peng-Robinson Equation of State with all BIPs set equal to zero. Take the temperature of the infinite environment to be equal to $T_0 = 300\text{ K}$

Power (kW)	13 94
Vapor fraction at the turbine outlet	12.03%
Lost Work (kW)	-392 X